

TAMPA KNIGHT, FL, USA

WMO: 720374

Lat: 27.916N

Lon: 82.449W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 92825

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.7	(c) 7.1	(d) -3.9	(e) 2.7	(f) 9.0	(g) -1.3	(h) 3.4	(i) 10.8	(j) 8.4	(k) 18.7	(l) 7.8	(m) 17.9	(n) 3.2	(o) 40	(p) 0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.4	(c) 35.0	(d) 26.0	(e) 33.9	(f) 25.8	(g) 32.9	(h) 25.5	(i) 28.0	(j) 31.5	(k) 27.7	(l) 31.1	(m) 27.4	(n) 30.6	(o) 2.9	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.3	29.2	27.1	22.9	29.1	26.3	21.8	28.8	89.9	30.9	87.7	32.0	86.0	30.3	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 7.1	(c) 6.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 8.1	(5) 7.1	(6) 6.0	(7) -0.5	(8) 28.1	(9) 2.4	(10) 1.8	(11) -2.2	(12) 29.5	(13) -3.6	(14) 30.5	(15) -4.9	(16) 31.6	(17) -6.7	(18) 32.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.7	16.0	18.1	20.4	24.0	26.6	28.4	28.7	29.0	28.2	24.7	20.6	19.8	(6)
(7)		DBStd	5.33	4.55	3.88	3.97	2.42	2.13	1.60	1.53	1.37	1.49	3.04	3.72	3.92	(7)
(8)		HDD10.0	11	9	2	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	231	97	48	28	1	0	0	0	0	0	4	21	33	(9)
(10)		CDD10.0	5024	194	228	322	421	516	551	581	588	547	455	317	303	(10)
(11)		CDD18.3	2204	24	41	92	172	258	301	323	329	297	201	88	78	(11)
(12)		CDH23.3	21079	94	178	578	1386	2529	3327	3624	3748	3138	1585	507	386	(12)
(13)		CDH26.7	8083	4	14	123	430	1042	1414	1546	1579	1273	498	101	58	(13)
(14)	Wind	WSAvg	2.8	2.8	2.7	3.0	3.1	2.9	2.5	2.1	2.4	2.8	3.1	3.0	2.8	(14)
(15)	Precipitation	PrecAvg	1323	66	67	82	65	73	205	207	222	187	60	40	58	(15)
(16)		PrecMax	1712	148	260	235	207	248	392	331	371	407	127	130	370	(16)
(17)		PrecMin	929	11	8	9	6	6	68	100	129	59	5	5	1	(17)
(18)		PrecStd	214	38	56	63	39	58	80	56	71	86	31	28	85	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.4	27.9	30.8	32.8	35.2	35.2	36.0	35.9	35.0	32.6	32.0	30.1	(19)
(20)			MCWB	21.2	20.9	21.9	22.3	23.9	25.6	26.5	26.8	25.8	25.3	25.1	23.7	(20)
(21)		2%	DB	26.1	27.1	29.1	31.4	33.0	34.0	34.1	34.1	33.7	32.1	29.0	27.9	(21)
(22)			MCWB	20.4	20.5	20.9	21.9	22.9	25.6	26.3	26.9	25.8	24.9	22.7	22.4	(22)
(23)		5%	DB	24.1	26.0	27.8	30.9	32.5	32.9	32.9	33.0	32.7	31.0	27.5	27.2	(23)
(24)			MCWB	19.8	19.9	20.5	22.1	22.9	25.3	26.4	26.5	25.8	24.2	21.6	22.0	(24)
(25)		10%	DB	22.6	24.0	27.1	29.1	32.0	32.4	32.5	32.5	32.3	29.8	26.3	26.0	(25)
(26)			MCWB	18.9	18.8	20.4	21.3	22.9	25.2	26.1	26.3	25.6	23.3	20.6	21.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	23.5	24.4	25.7	26.7	28.6	28.5	28.6	28.2	27.2	25.8	24.9	(27)
(28)			MCD B	25.7	26.6	28.1	29.7	30.0	32.2	32.2	32.4	31.2	29.8	29.1	27.9	(28)
(29)		2%	WB	22.2	22.5	23.5	24.8	25.7	27.6	27.9	28.0	27.5	26.4	24.6	23.9	(29)
(30)			MCD B	24.0	25.3	26.7	28.7	29.6	31.4	31.4	31.6	30.6	28.6	27.8	27.0	(30)
(31)		5%	WB	21.0	21.4	22.6	24.0	25.2	27.1	27.5	27.7	27.2	25.7	23.3	23.2	(31)
(32)			MCD B	22.8	23.6	25.5	28.0	29.4	30.7	30.8	31.1	30.2	28.5	26.2	25.8	(32)
(33)		10%	WB	19.7	20.6	21.7	23.1	24.6	26.6	27.1	27.4	26.7	25.0	22.3	22.4	(33)
(34)			MCD B	22.7	22.6	24.7	26.8	29.0	29.9	30.4	30.5	29.7	28.1	24.9	24.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.0	10.2	9.9	9.7	8.0	7.5	7.4	8.1	8.8	9.8	9.3	(35)
(36)			MCDBR	10.3	10.1	10.6	10.5	10.5	8.9	8.2	8.2	8.4	9.0	9.6	9.2	(36)
(37)			MCWBR	5.7	5.3	4.2	3.5	3.2	3.2	3.1	3.0	3.0	3.4	4.4	4.6	(37)
(38)		5% WB	MCDBR	8.2	8.5	8.5	8.6	8.7	7.6	7.3	7.5	7.6	7.7	8.0	(38)	
(39)			MCWBR	6.0	5.8	4.7	4.5	3.3	3.4	3.3	3.2	3.1	3.3	4.3	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.355	0.368	0.391	0.425	0.450	0.474	0.455	0.422	0.390	0.354	0.342	(40)	
(41)		taud	2.520	2.493	2.443	2.343	2.307	2.316	2.252	2.321	2.382	2.454	2.499	2.547	(41)	
(42)		Ebn at Noon	903	919	923	906	870	843	822	838	858	869	881	887	(42)	
(43)		Edh at Noon	91	101	112	127	132	130	139	129	118	103	92	85	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.54	4.28	5.40	6.46	6.84	6.16	5.84	5.63	5.13	4.73	3.86	3.28	(44)	
(45)		RadStd	0.25	0.27	0.44	0.31	0.38	0.50	0.36	0.35	0.33	0.29	0.20	0.27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	-4	-47	+164	+116

Nomenclature: See separate page